Softmax Inundation and Generational Context in Long-Horizon Agent Systems

Analytical Attention Dilution Bounds and the Generational Memory Architecture

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Preprint Ref: ACI-MATH-2026-FT-REV2 • October 2026

Abstract. *Autonomous AI agent architectures routinely treat the transformer context window as an unbounded, append-only linear interaction transcript. We analyze the theoretical and empirical consequences of this paradigm. We formalize the* ***Softmax Inundation Bound****, an analytical upper bound showing that under bounded projection energy, the maximum aggregate self-attention probability mass allocated to a fixed set of invariant constraint tokens decays asymptotically as O(L_u / |S_noise|) when unpruned execution observations accumulate. This attention dilution creates a systemic vulnerability: as noisy tracebacks dominate context length, attention heads dilute focus from foundational system invariants toward high-entropy symptom tokens. To resolve this failure mode, we specify* ***Generational Context****, an operating-system-inspired memory architecture that separates memory into* ***Gen-0 Ephemeral*** *(tool outputs distilled to 3-line signatures and evicted in 1 turn),* ***Gen-1 Evaporative*** *(working memory paged via the Attention-Weighted Eviction algorithm), and* ***Gen-2 Tenured*** *(permanently pinned invariants). We prove that Generational Context maintains a non-zero lower bound on invariant attention uniformly for all interaction turns t ∈ N, eliminating unbounded softmax dilution and stabilizing long-horizon autonomy.*

1. Introduction

Despite expansions in foundation model context windows, practical autonomy in multi-turn software engineering loops remains fragile. Agents frequently lose adherence to system constraints, repeat failed tool invocations, and apply superficial micro-patches around symptoms rather than addressing root causes.

In typical agent sessions, over 90% of active tokens consist of raw tool outputs (compiler stderr, test dumps, DOM trees). We analyze this pathology via **Payload Inundation** (attention dilution), **Error Gravitation** (semantic bias toward error tracebacks), and **Generational Context** (multi-tiered memory management).

2. The Softmax Inundation Bound

Let context tokens be partitioned into Invariant Subspace S_inv (|S_inv| = L_u) and Observation Subspace S_noise. Under bounded query and key energies ||q_i||_2 ≤ M_Q and ||k_j||_2 ≤ M_K, let κ = exp(2 M_Q M_K / sqrt(d_k)).

Theorem 1 (Softmax Inundation Upper Bound).
For any query position i, the aggregate attention probability mass allocated to the invariant subspace S_inv is upper-bounded by:

$$A_{i,inv} \le \kappa \frac{L_u}{L_u + |S_{noise}|} = O\left(\frac{L_u}{|S_{noise}|}\right)$$

Proof Sketch: By Cauchy-Schwarz, -M_Q M_K ≤ q_i^T k_j ≤ M_Q M_K. The numerator is bounded above by L_u exp(M_Q M_K / sqrt(d_k)) and the denominator is bounded below by (L_u + |S_noise|) exp(-M_Q M_K / sqrt(d_k)). Dividing numerator by denominator yields the result directly.

3. Generational Context & Retention Guarantee

Generational Context stratifies memory into three lifecycle tiers: **Gen-0 Ephemeral** (tool outputs distilled to 3 lines and purged in 1 turn), **Gen-1 Evaporative** (working memory paged via AWE), and **Gen-2 Tenured** (permanently pinned invariants).

Theorem 2 (Generational Retention Guarantee).
Under Generational Context with bounded distillation $||\sigma_distill(G_0)|| \leq K_0$ *and working capacity* $||G_1|| \leq C_1$*, the active non-invariant tokens satisfy* $|S_non\text{-}inv^{(t)}| \leq C_1 + K_0 = C_bound$ *for all turns t. The invariant attention is strictly lower-bounded:*

$$A_{i, inv} \geq L_u / (L_u + \kappa C_{bound}) > 0$$

Invariant attention remains strictly bounded away from zero uniformly for all interaction turns $t \in N$.

4. The Attention-Weighted Eviction (AWE) Algorithm

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Algorithm 1: AWE with Hard Capacity Invariant
  1: Input: Capacity C, incoming chunk E, boundary parameter p
  2: if E.tokens > C then Truncate E.content to C tokens; E.tokens ← C
  3: Compute IDS(E) = AttentionWeight(E) × 100 / (Entropy(E) + 10^-4) [or tier prior if unmeasured]
  4: if E.key ∈ T1 ∪ T2 then Update and promote to T2
  5: else if E.key ∈ B1 then Adapt p ← min(C, p + |B2|/|B1| × E.tokens); Evict lowest IDS victim
  6: else if E.key ∈ B2 then Adapt p ← max(0, p - |B1|/|B2| × E.tokens); Evict lowest IDS victim
  7: else Evict lowest IDS chunk; Insert E into T1; assert total_tokens() ≤ C
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5. Empirical Token Compaction Results

Architecture Arm	Mean Active Tokens	Context Signal Ratio (CSR)	Token Reduction
Arm A: Flat Append-Only	5,171	0.0057	Baseline (0.0%)
Arm B: Auto-Compaction	3,400	0.0076	-34.2%
Arm C: Generational Context	140	0.2375	**-97.3% (41x CSR)**

6. Related Work & Conclusion

Our work connects with Liu et al. (2023) on context retrieval degradation, Xiao et al. (2023) on attention sinks, Packer et al. (2023) on MemGPT tiered memory, and Megiddo & Modha (2003) on ARC. By formalizing the Softmax Inundation Bound and providing the Generational Context architecture, we offer a deterministic, verifiable memory framework that preserves foundational constraints across arbitrary interaction horizons.

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